

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

COMBINED SCIENCE

4003/2

PAPER 2 Theory

NOVEMBER 2020 SESSION

2 hours

Additional materials:
Answer sheets
Calculator (Optional)
String

The Periodic Table is provided on page 15.

Time 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

Section A

Answer **all** questions.

write your answers in the spaces provided on the question paper.

Section B

Answer any **two** questions.

Write your answers on the separate answer paper provided.

Section C

Answer any **two** questions.

Write your answers on the separate sheets provided.

Section D

Answer any **two** questions.

Write your answers on the separate answer sheets provided.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

For examiner's use

Section A

Section B

Section C

Section D

TOTAL

This question paper consists of 16 printed pages.
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Section A

*Answer **all** the questions in the spaces provided.*

1. (a)

Define the term *ecosystem*.

.....

.....

.....

[2]

(b)

State any **two** physical components of an ecosystem.

1

.....

2

.....

[2]

(c)

Draw a food chain, with three trophic levels, involving **named** organisms

[3]

2. **Fig.2.1** shows a cell membrane separating water and sugar solution.

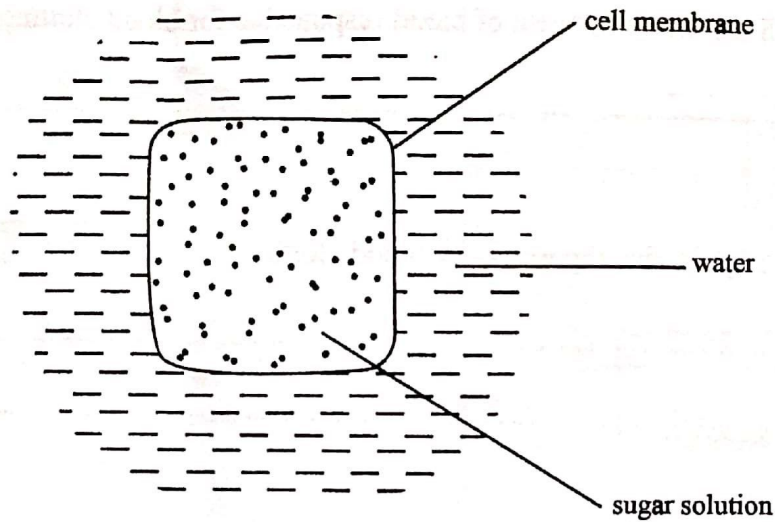


Fig.2.1

- (a) (i) Explain how osmosis occurs.

.....

.....

.....

[2]

- (ii) Define the term *diffusion*.

.....

.....

[1]

- (b) Blood circulating around the body has plasma, water, platelets, glucose, red blood cells, carbon dioxide, oxygen and white blood cells.

- (i) Select **two** substances that are needed by respiring cells.

1

.....

2

.....

[2]

- (ii) State the component of blood responsible for blood clotting.

.....

[1]

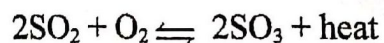
- (c) Explain the importance of blood clotting.

.....

.....

[1]

3. Sulphur dioxide (SO₂) reacts with oxygen (O₂) to form sulphur trioxide (SO₃) during the manufacture of sulphuric acid as shown in the equation:



- (a) State the oxidation state of sulphur in sulphur trioxide.

.....

[1]

- (b) State the meaning of the symbol $\rightleftharpoons$.

.....

[1]

- (c) Write an equation to show the formation of oleum.

.....

[2]

- (d) State any two industrial uses of sulphuric acid.

4

Fig.4.1 shows the stages involved in the extraction of gases A and B from air.

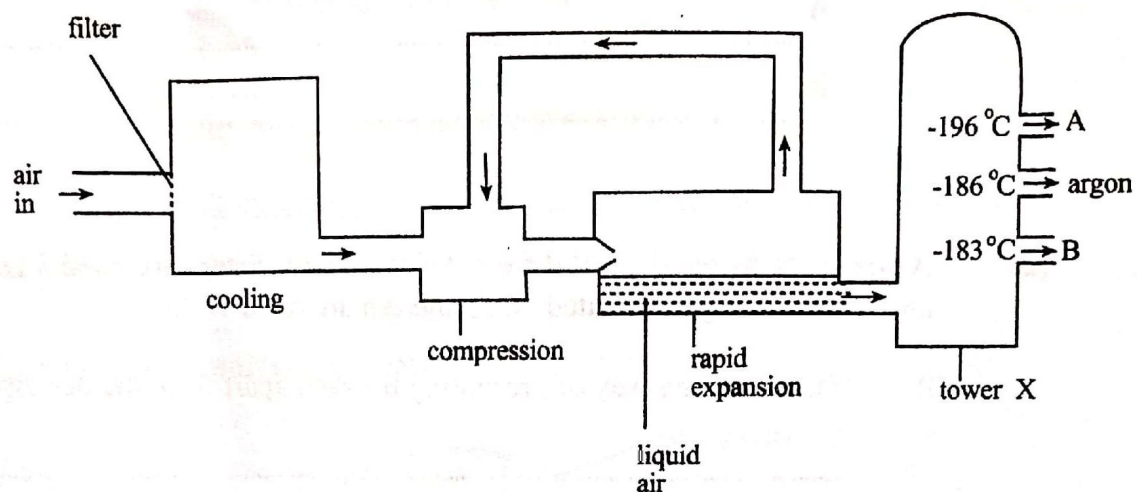


Fig.4.1

- (a) Name the component that is removed during filtration.

.....

[1]

- (b) (i) Identify the process that occurs in tower X.

.....

[1]

- (ii) Explain why the air is compressed and expanded rapidly and repeatedly.

.....

[2]

- (c) Explain why gas A is collected first.

.....

[1]

(d) Identify the gases A and B.

A

.....

B

.....

[2]

5. (a) A box contains resistors of different sizes. Ten resistors are rated $5\ \Omega$, twelve are rated $8\ \Omega$, eight are rated $10\ \Omega$ and ten are rated $15\ \Omega$.

(i) State any **one** way of presenting the data apart from the description.

.....

[1]

(ii) Calculate the percentage of the resistors rated $15\ \Omega$.

[2]

(b) One side of a cube measures 6.0 cm.

Calculate the volume of the cube.

[2]

(c) Give the most suitable instrument for measuring

(i) the internal diameter of a test tube,

(ii) the external diameter of a wire.

[2]

6. Fig.6.1 shows a simple solar cooker.

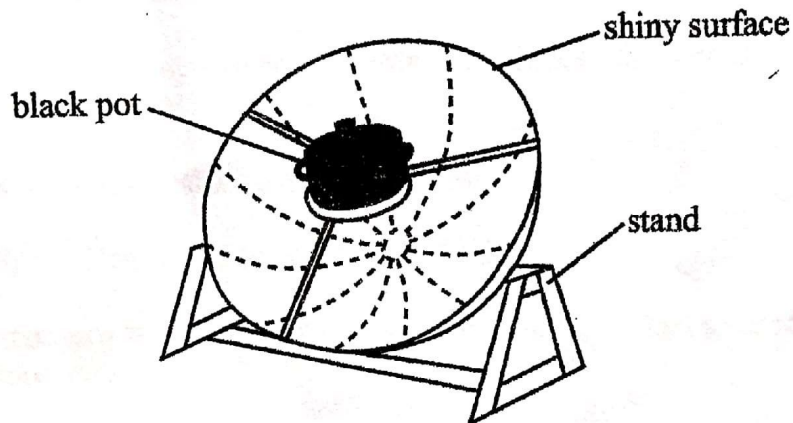


Fig.6.1

(a) (i) Explain the function of the shiny curved surface.

[2]

(ii) Explain why the pot is painted black.

[1]

- (b) (i) Give a reason why solar energy is called a renewable source of energy.

.....
[1]

- (ii) Explain how solar energy is indirectly used to produce electricity.

.....
.....
[2]

Section B

Answer any two questions on the separate answer sheets provided.

7. (a) (i) State any **three** methods of contraception. [3]
- (ii) Describe how each method stated in (i) prevents fertilisation. [3]
- (b) (i) Define the term *rhizome*. [1]
- (ii) State any **three** advantages of reproducing plants by means of rhizomes. [3]
8. (a) (i) State any **two** components of a balanced diet and state **one** function for each. [4]
- (ii) Explain the term *deficiency disease*. [1]
- (iii) Give any **three** deficiency diseases. [3]
- (b) State any **two** processes through which energy is lost in food chains and food webs. [2]
9. (a) (i) Define the term *immunity*. [1]
- (ii) State the **two** methods by which infants acquire immunity. [2]
- (b) Describe the process of vaccination. [4]
- (c) Describe any **three** methods of controlling mosquitoes. [3]

Section C

Answer any two questions on the separate answer sheets provided.

10. (a) The reaction between copper oxide (CuO) and hydrogen (H₂) involves oxidation and reduction.



- (i) Define the term oxidation. [1]
- (ii) Identify the reducing agent in the reaction between copper oxide and hydrogen. [1]
- (b) Iron is extracted from iron(III) oxide in the blast furnace.
- (i) Name any **two** raw materials that are fed into the furnace from the top. [2]
- (ii) Write a chemical equation for the reaction that produces iron. [2]
- (c) (i) Describe, with the help of equations, how slag (calcium silicate) is formed in the blast furnace. [3]
- (ii) State any **one** metal which is added to iron to make stainless steel. [1]

11. (a) Fig.11.1 represents an incomplete electrolytic cell that can be used for copper plating an iron object.

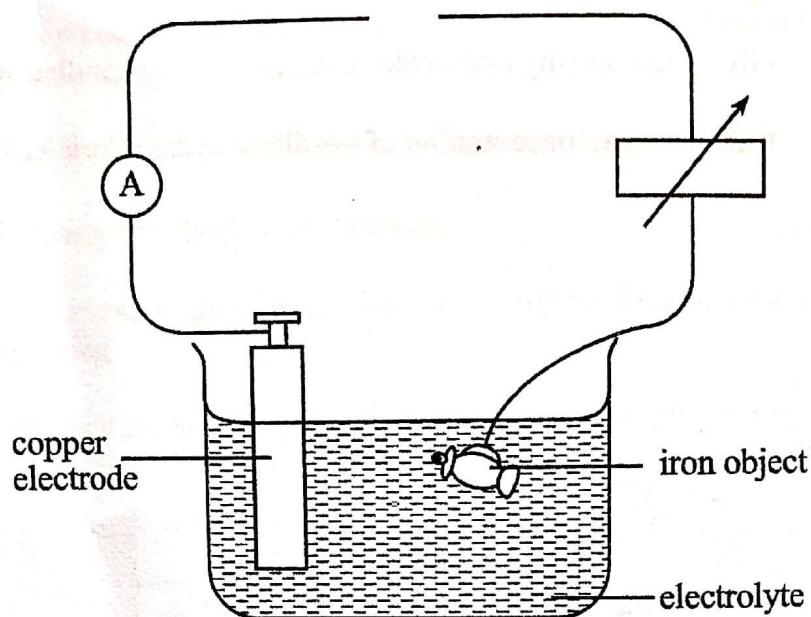


Fig.11.1

- (i) Define the term *electrolysis*. [1]
- (ii) Name the electrolyte used in the process. [1]
- (iii) Identify the missing apparatus. [1]
- (b) State **one** change which occurs at the
- (i) copper electrode, [1]
- (ii) iron object. [1]
- (c) State **two** reasons for electroplating iron objects. [2]
- (d) Write the half equation for the copper electrode. [3]

12. A volume of 0.025 dm^3 of 0.1 mol/dm^3 sodium hydroxide (NaOH) solution was put into a flask and a few drops of phenolphthalein indicator were added. A volume of 0.020 dm^3 of dilute hydrochloric acid (HCl) was used to completely neutralise sodium hydroxide solution.

- (a) Name the method of neutralisation used. [1]

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[Turn over]

- (b) Explain the purpose of adding phenolphthalein indicator. [1]
- (c) Write a chemical equation for the reaction between sodium hydroxide and hydrochloric acid. [2]
- (d) Give a reason why hydrochloric acid should be handled with care. [1]
- (e) Calculate the concentration of the dilute hydrochloric acid. [5]

Section D

Answer any two questions on the separate answer sheets provided.

13. (a) Explain how a positively charged gold leaf electroscope can be used to test the charge on a charged rod. [4]
- (b) (i) Describe how a house is protected by a lightning conductor. [3]
- (ii) State any two dangers of lightning. [2]
- (iii) State any one safety precaution that should be taken against lightning. [1]

14. (a) Fig.14.1 shows a model of a hydroelectric power station designed by a student.

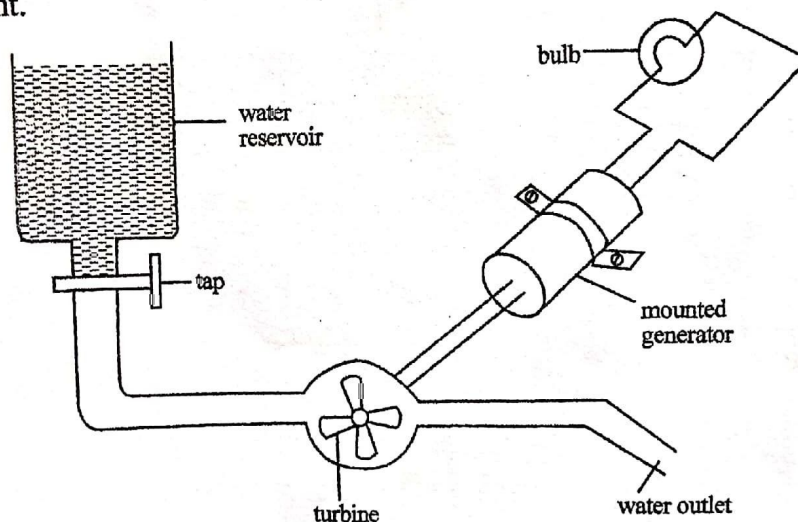


Fig.14.1

- (i) Describe, using energy changes, how the bulb in the model lights. [3]
- (ii) Explain why using a hydroelectric power station is better than using a thermal power station. [2]
- (b) (i) State the relationship between voltage and current. [1]
- (ii) State any two factors that affect resistance of an electrical conductor. [2]
- (iii) Calculate the current flowing in a circuit if the potential difference across a $2\ \Omega$ resistor is 6 V. [2]

15. (a) A pot with water is placed on an electric stove and the water is heated until it boils.
- (i) State the modes of heat transfer involved. [2]
 - (ii) Describe the modes of heat transfer stated in (a)(i). [3]
- (b) The electric stove is marked 2 000 W.
- (i) Define the terms *electrical power* and *electrical energy*. [2]
 - (ii) The water in the pot boils after 8 minutes.
Calculate the energy supplied by the electric stove. [3]

DATA SHEET

The Periodic Table of the Elements

Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	O		
7 Li Lithium 3 9 Be Beryllium 4 23 Na Sodium 11 24 Mg Magnesium 12 39 K Potassium 19 40 Ca Calcium 20 85 Rb Rubidium 37 133 Cs Cesium 55 87 Fr Francium 87		1 H Hydrogen 1												11 B Boron 5 27 Al Aluminium 13 70 Ga Gallium 31 65 Zn Zinc 30 115 In Indium 49 204 Tl Thallium 81	12 C Carbon 6 28 Si Silicon 14 73 Ge Germanium 32 119 Sn Tin 50 207 Pb Lead 82	14 N Nitrogen 7 31 P Phosphorus 15 75 As Arsenic 33 122 Sb Antimony 51 209 Bi Bismuth 83	16 O Oxygen 8 32 S Sulphur 16 79 Se Selenium 34 128 Te Tellurium 52 210 Po Polonium 84	19 F Fluorine 9 35.5 Cl Chlorine 17 80 Br Bromine 35 127 I Iodine 53 210 At Astatine 85	4 He Helium 2 20 Ne Neon 10 40 Ar Argon 18 84 Kr Krypton 36 131 Xe Xenon 54 222 Rn Radon 86
140 Ce Cerium 58 141 Pr Praseodymium 59 144 Nd Neodymium 60 150 Sm Samarium 62 152 Eu Europium 63 157 Gd Gadolinium 64 159 Tb Terbium 65 162 Dy Dysprosium 66 165 Ho Holmium 67 167 Er Erbium 68 173 Yb Ytterbium 70 175 Lu Lutetium 71																			
232 Th Thorium 90 232 Pa Protactinium 91 238 U Uranium 92 238 Np Neptunium 93 238 Pu Plutonium 94 238 Am Americium 95 238 Cm Curium 96 238 Bk Berkelium 97 238 Cf Californium 98 238 Es Einsteinium 99 238 Fm Fermium 100 238 Md Mendelevium 101 238 No Nobelium 102 238 Lr Lawrencium 103																			

a

X

b

a = relative atomic mass

X = atomic symbol

b = proton (atomic) Number

*58-71 Lanthanoid series
 190-103 Actinoid series

Key

a	X	b
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a = relative atomic mass
X = atomic symbol
b = proton (atomic) Number

The volume of one mole of any gas is 28 dm³ at room temperature and pressure (r.t.p.)